

M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 9 (9TH MATRIC) - MATHEMATICS (SCIENCE GROUP)

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Science Group | Total Marks: 75 Marks (Theory: 75) | Time: 2 Hours 30 Minutes

SECTION 1: BISE GUJRWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- Section A: Objective (MCQs):

15 Compulsory MCQs (1 Mark each = 15 Marks). 1-2 questions from each chapter.

- Section B: Short Questions:

Attempt 18 out of 27 Short Questions (3 sub-sections: Q2, Q3, Q4: 6/9 each = 36 Marks).

- Section C: Long Questions:

Attempt 3 out of 5 Long Questions (8 Marks each = 24 Marks). Question 9 (Theorems from Ch 12 or Ch 16) is COMPULSORY.

- Mandatory Theorem Requirement:

Q9 has 8 marks dedicated to proving one geometry theorem from Chapter 12 or Chapter 16.

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Chapter 1: Matrices and Determinants:

Order of matrix, Types of matrices (Row, Column, Square, Rectangular, Zero, Identity, Diagonal, Scalar), Transpose, Symmetric vs Skew-Symmetric, Matrix addition and multiplication condition, Determinant of 2×2 matrix, Singular vs Non-singular, Adjoint of matrix, Multiplicative Inverse ($A^{-1} = \frac{1}{|A|} \cdot \text{adj } A$), Solving system of linear equations: Matrix Inversion Method and Cramer Rule.

Chapter 2: Real and Complex Numbers:

Real numbers classification (Rational vs Irrational), Radicals and Radicands, Laws of Exponents ($x^a \cdot x^b = x^{(a+b)}$, $(x^a)^b = x^{(ab)}$), Complex numbers ($z = a + bi$, $i = \sqrt{-1}$, $i^2 = -1$), Conjugate of complex number, Basic operations on complex numbers (+, -, *, /).

Chapter 3: Logarithms:

Scientific notation, Definition of Logarithm ($y = \log_b x \Leftrightarrow b^y = x$), Common (Base 10) vs Natural (Base e) logarithms, Characteristic and Mantissa, Laws of Logarithms: $\log(mn) = \log m + \log n$; $\log(m/n) = \log m - \log n$; $\log(m^n) = n \log m$; Change of base rule; Application of logarithms in calculating complex numerical expressions.

Chapter 4: Algebraic Expressions & Algebraic Formulas:

Polynomials and Degree, Rational expressions, Value of algebraic expression, Core Formulas: $(a+b)^2$, $(a-b)^2$, $a^2 - b^2$, $(a+b+c)^2$, $(a+b)^3$, $(a-b)^3$, $a^3 + b^3$, $a^3 - b^3$, Surds and their conjugate (rationalization of denominator).

Chapter 5: Factorization:

Techniques of factorization: Common terms, Grouping, Using standard formulas, Completing the square, Mid-term break method, Remainder Theorem and Factor Theorem, Factorization of cubic polynomials.

Chapter 6: Algebraic Manipulation:

Highest Common Factor (HCF) by Factorization and Division methods, Least Common Multiple (LCM), Relation between HCF and LCM: $\text{HCF} \times \text{LCM} = p(x) \times q(x)$, Square root of algebraic expressions by factorization and division.

Chapter 7: Linear Equations & Inequalities:

Linear equation in one variable, Equations involving absolute value ($|x| = a \Rightarrow x = \pm a$), Extraneous roots, Linear inequalities and their solution sets on real number line.

Chapter 9: Introduction to Coordinate Geometry:

Distance Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, Collinear points condition, Types of triangles (Equilateral, Isosceles, Scalene, Right-angled using Pythagoras theorem), Mid-point Formula: $R = ((x_1+x_2)/2, (y_1+y_2)/2)$.

Chapter 12: Line Bisectors and Angle Bisectors (Theorems):

Theorem 12.1: Any point on the right bisector of a line segment is equidistant from its end points. Theorem 12.2: Any point equidistant from end points lies on right bisector. Theorem 12.4: Any point on bisector of an angle is equidistant from its arms.

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

- * Question 9 Theorem is MANDATORY: Always write Given, To Prove, Construction, and Table of Statements & Reasons with an accurate diagram.
- * Show every step in Cramer Rule and Matrix Inversion method (Determinant, Adjoint, Inverse, Multiplication).
- * For logarithm calculations, state the law of logarithm used before applying anti-logarithm.